UNIT 3 · LESSON 2 · LT 3.3 · BIO TOOL #11 · ADVANCED

Turn Off the Oxygen

The Energy Patch · Module 2, Mitochondrion

Name _____ Period ____ Date _____

Open **Bio Tool #11, the Energy Patch**. Set **Running on: iPad**, then tap **2 · Mitochondrion**. Tap whatever wears the green ring. The driver taps and the reader circles; swap jobs at step 3.
scienceonepagers.org/biology/tools/organelles-energy.html

Step 1 · The glucose ring breaks in the cytoplasm

1. As the glucose ring breaks in the cytoplasm, how many carbon dioxide molecules leave?
Ⓐ 0 Ⓑ 1 Ⓒ 2 Ⓓ 6

Step 2 · A three-carbon piece goes in

2. Inside the mitochondrion one three-carbon piece is trimmed. What leaves, and how many carbons stay in the piece?
Ⓐ O_2; 3 stay Ⓑ CO_2; 2 stay Ⓒ CO_2; 3 stay Ⓓ two CO_2; 1 stays

Step 3 · Count every carbon dioxide (swap jobs)

3. Read the carbon count when one whole glucose is done.
Ⓐ 6 in, 3 out, 3 still inside
Ⓑ 6 in, 12 out as CO_2
Ⓒ 6 in, 0 out; the carbons became energy
Ⓓ 6 in, 6 out as CO_2, 0 inside

Step 4 · Oxygen keeps the line running

4. Oxygen on. Feed one glucose. The ATP tally ends near:
Ⓐ 2 Ⓑ 6 Ⓒ about 30 or more Ⓓ over 100

5. Turn the oxygen off. Which **two** stop coming out of the mitochondrion? Circle two.
Ⓐ carbon dioxide Ⓑ ATP Ⓒ glucose Ⓓ chlorophyll Ⓔ oxygen

6. Oxygen still off. Feed one glucose. The tally ends at:
Ⓐ 0
Ⓑ 2, made in the cytoplasm
Ⓒ 2, made in the mitochondrion
Ⓓ about 32

7. Under **Which cell?** try both. Which pairing matches the screen?
Ⓐ yeast or bacteria → lactic acid; muscle → ethanol + CO_2
Ⓑ both make lactic acid
Ⓒ yeast or bacteria → ethanol + CO_2; muscle → lactic acid
Ⓓ both make ethanol + CO_2

At the bag reading

8. The bag with sugar puffed up; the yeast-and-water bag stayed flat. Which explanation fits the bags and the screen?
Ⓐ Oxygen: yeast give off oxygen as they eat sugar
Ⓑ Carbon dioxide: with little oxygen left in the sealed bag each sugar gives only 2 ATP, so the yeast go through a lot of sugar
Ⓒ Carbon dioxide: fermentation makes more ATP than oxygen does
Ⓓ Ethanol vapor: the alcohol boils off

The Energy Patch's model lands near 32 ATP per glucose with oxygen. Slides and the test round to about 36. Either way it is far more than the 2 a cell keeps without oxygen.

Optional, not scored: Why does turning off the oxygen stop ATP synthase? Use **electrons** and **hydrogen ions**.

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UNIT 3 · LESSON 2 · LT 3.3 · BIO TOOL #11 · CORE

What Happens Without Oxygen

The Energy Patch · Module 2, Mitochondrion

Name _____ Period ____ Date _____

Watch the board while your teacher runs steps 1 to 3. At step 4 you tap two things on your iPad: the **Oxygen** switch and **Which cell?**

scienceonepagers.org/biology/tools/organelles-energy.html

Step 3 · Count the carbon

1. One whole glucose is done. Read the carbon count.

(A) 6 in, 0 out (B) 6 in, 3 out (C) 6 in, 6 out as CO_2

Step 4 · Oxygen on, then off

2. Oxygen on. One glucose gives the cell:

(A) a little ATP (B) far more ATP than without oxygen (C) no ATP

3. Oxygen off. Circle the **two** that stop coming out of the mitochondrion.

(A) carbon dioxide (B) ATP (C) glucose (D) oxygen

4. Which cell? Match each cell to what it makes without oxygen.

(A) yeast → ethanol + CO_2, muscle → lactic acid
(B) yeast → lactic acid, muscle → ethanol + CO_2

At the bag reading

5. The bag with sugar puffed up. Which gas filled it?

(A) oxygen (B) carbon dioxide (C) glucose

With oxygen a cell gets far more ATP from each sugar. Without oxygen it keeps only 2.

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